BCSE316L | DESIGN OF SMART CITIES

Citizen Sentiment Analysis of Municipal Services Using Transformer-Based NLP

Review-2 / Final Review Presentation

Vellore Institute of Technology • SCOPE

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Project / Team 16

October 2026

A reproducible civic intelligence pipeline from open data to ~~decision support~~

1.20M

311 service requests

406.6K

RSAT responses

EXECUTIVE STORY

NLP adds evidence beyond traditional operational KPIs

Core question

Does adding sentiment, topic, and grievance signals improve out-of-time prediction of resolution delay and repeat complaints?

The answer is target-specific: the corrected experiment supports a significant delay gain, while repeat-rate improvement remains an observed but uncertain signal.

+0.0289
Delay ΔR^2 | p = 0.00020

+0.0257
Repeat ΔR^2 | p = 0.733

16 / 16
Current automated tests passing

884
Integrated cells

What is ready for review

- Audited multi-source data pipeline
- VADER, LDA, BERTopic, ABSA, and DistilBERT experiments
- Chronological model comparison with robustness analysis
- Two-sided Streamlit dashboard for operations and residents

Administrative closure does not equal citizen satisfaction

Traditional KPI

- Ticket volume
- Open-to-close time
- Backlog and staffing
- Closed status

gap

Citizen reality

- Was the issue corrected?
- Did the request recur?
- Was communication adequate?
- Was the work satisfactory?

Research opportunity Use public survey ground truth and operational narratives to test whether citizen-perceived friction contains incremental predictive information.

RESEARCH DESIGN

The study tests predictive value, not causality

Research question — **Does augmenting structured operational indicators with NLP sentiment, topic, and aspect signals improve out-of-time prediction?**

Hypotheses

H1 | Delay

NLP-augmented features improve prediction of resolution delay.

H2 | Repeat rate

NLP-augmented features improve prediction of 14-day repeat requests.

Interpretation rule — All findings are aggregate predictive associations at service × borough × month grain.

Two authentic NYC Open Data sources create complementary evidence

NYC 311 service requests

Socrata ID: erm2-nwe9

1,199,279 records

- Operational closure narratives
- Created and closed timestamps
- Delay and repeat-request outcomes
- Top 20 service categories

Resolution Satisfaction Survey

Socrata ID: 5ijn-vbdv

406,581 responses

- 5-point satisfaction ground truth
- Standardized dissatisfaction reasons
- Borough and month fields
- Public free-text suppressed for PII

Audit decision NLP is extracted from NYC 311 resolution narratives; RSAT supplies survey-level labels and satisfaction aggregates.

The join is privacy-safe and the evaluation is chronological

Robust cell grain

c = (service, borough, year, month)

No one-to-one survey-to-ticket join is claimed.

Chronological partitions

TRAIN
2023 Jan–Aug
586 cells

VALIDATION
2023 Sep–Dec
213 cells

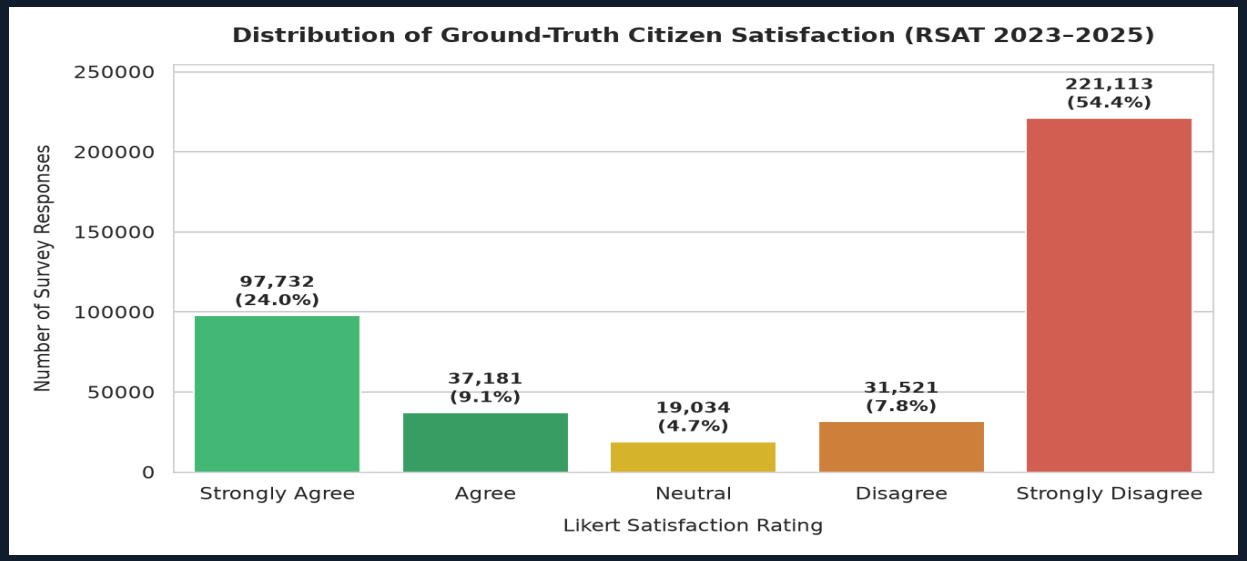
OUT-OF-TIME TEST
2024 horizon
85 cells

The survey reveals a strongly negative satisfaction distribution

62.14%

of validated RSAT responses are dissatisfied

Issue correction is the largest standardized grievance aspect (42.7% of dissatisfied responses).



RSAT satisfaction distribution This skew establishes a meaningful citizen-distress signal for later aggregation and modeling.

NLP METHODOLOGY

The NLP stack moves from baselines to domain adaptation

VADER

Lexicon sentiment

Accuracy 0.655
Negative F1 0.800

LDA + BERTopic

Themes and clusters

C_v 0.5100
14 semantic clusters

ABSA

Six grievance aspects

TF-IDF macro-F1 0.1723
Rule baseline 0.1295

DistilBERT

Transformer fine-tuning

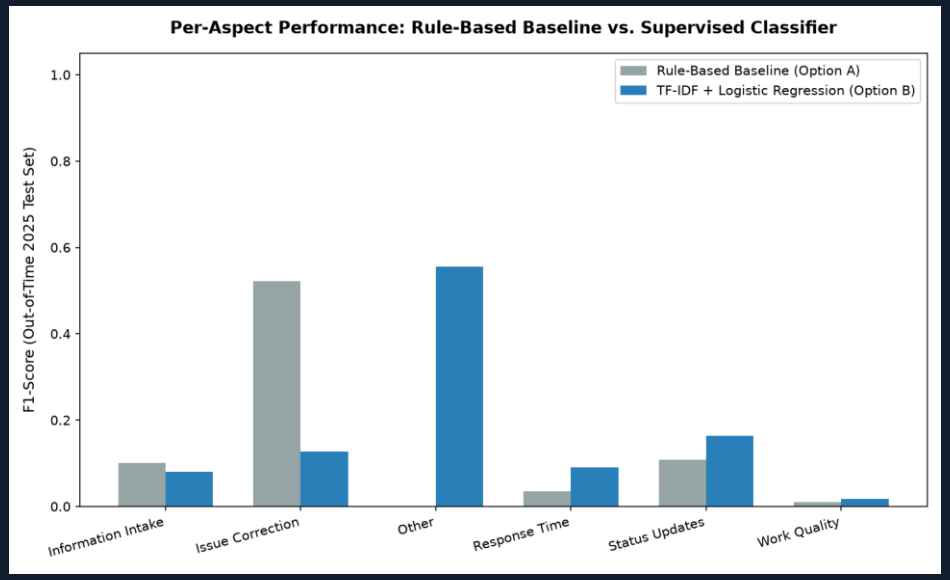
Test macro-F1 0.3491
Accuracy 0.4148

Design boundary

Resolution narratives are agency text; RSAT provides the citizen-grounded labels and satisfaction aggregates.

NLP RESULTS

Transformer adaptation improves grievance attribution



Per-aspect F1 comparison

Transformer run

0.3491

Out-of-time macro-F1

+0.1768

Macro-F1 gain vs TF-IDF

Kaggle Tesla T4 • 3 epochs • 2e-5 learning rate
41,746 chronological examples across train / validation / test

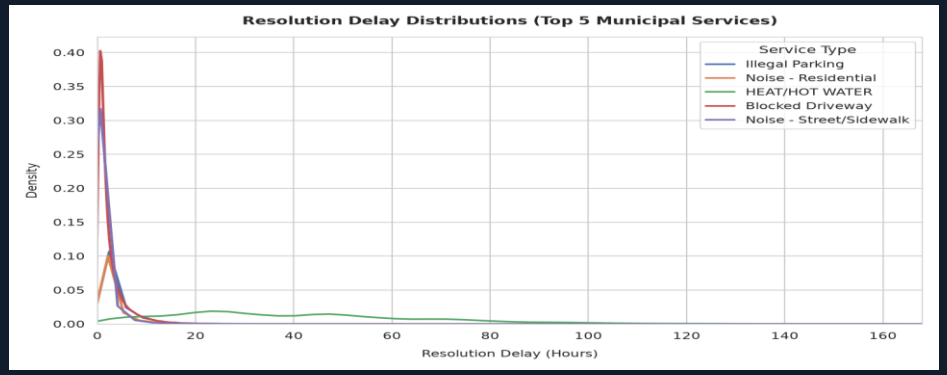
Operational outcomes quantify both delay and recurring demand

2.90 h
Median raw resolution delay

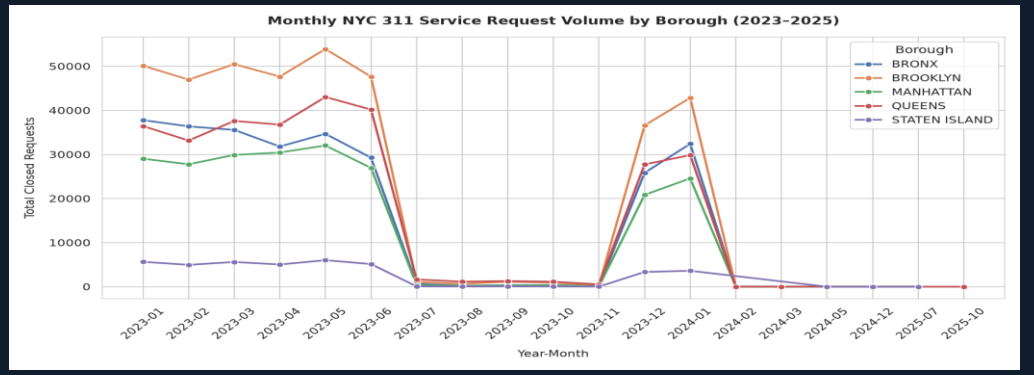
91.11 h
Mean winsorized delay

44.88%
14-day repeat rate

50.52%
30-day repeat rate



Delay is highly right-skewed; p99 winsorization cutoff = 2,454.96 hours.



Repeat rates rise with longer windows: 39.10% at 7 days → 50.52% at 30 days.

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The experiment isolates NLP's incremental value

Structured baseline

- Service identity
- Borough
- Month seasonality
- Request volume

→

NLP-augmented model

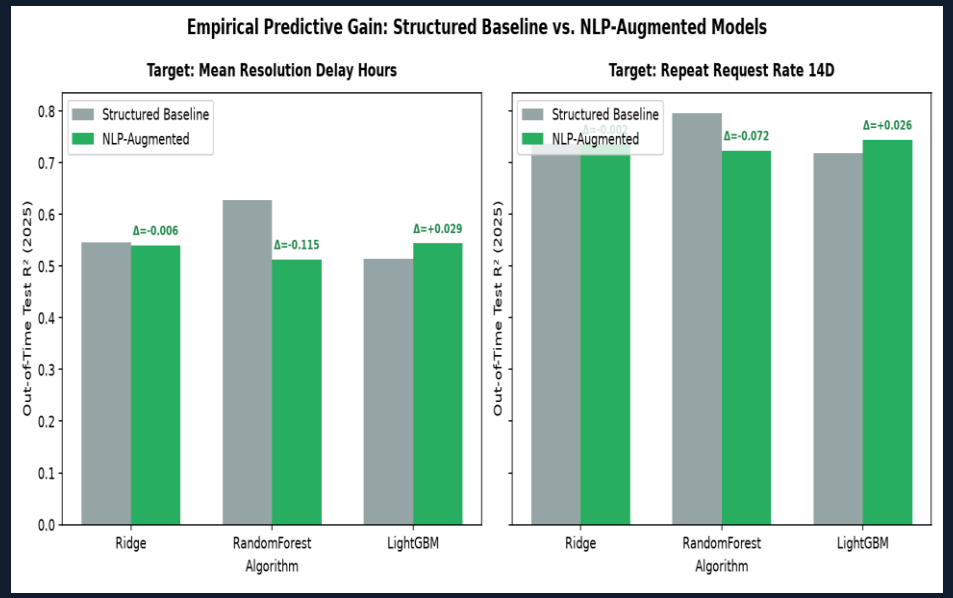
- All baseline features
- VADER sentiment rates
- LDA / BERTopic features
- ABSA aspect proportions
- RSAT satisfaction signals

Evaluation protocol

Ridge • Random Forest • LightGBM | validation-selected algorithm | one untouched 2024 horizon test

MAIN RESULTS

Delay gains are significant; repeat-rate gains are inconclusive



2024 out-of-time R^2 comparison

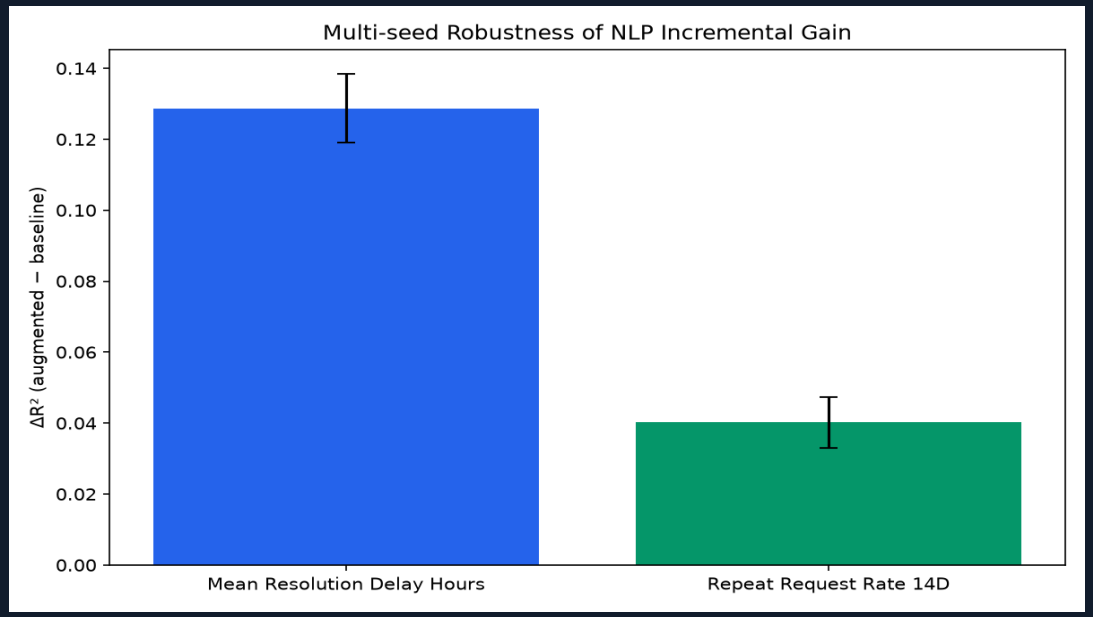
Target	Model	Base R^2	Aug R^2	ΔR^2	p-value
Delay (h)	LightGBM	0.5149	0.5438	**+0.0289**	0.00020
Repeat 14d	LightGBM	0.7184	0.7441	**+0.0257**	0.733

Interpretation

Delay: confirmed incremental predictive association.
Repeat: positive observed gain, but not statistically significant.

UNCERTAINTY ANALYSIS

Robustness confirms delay gains and qualifies repeat-rate gains



Five seeds + 2,000 bootstrap resamples

Delay

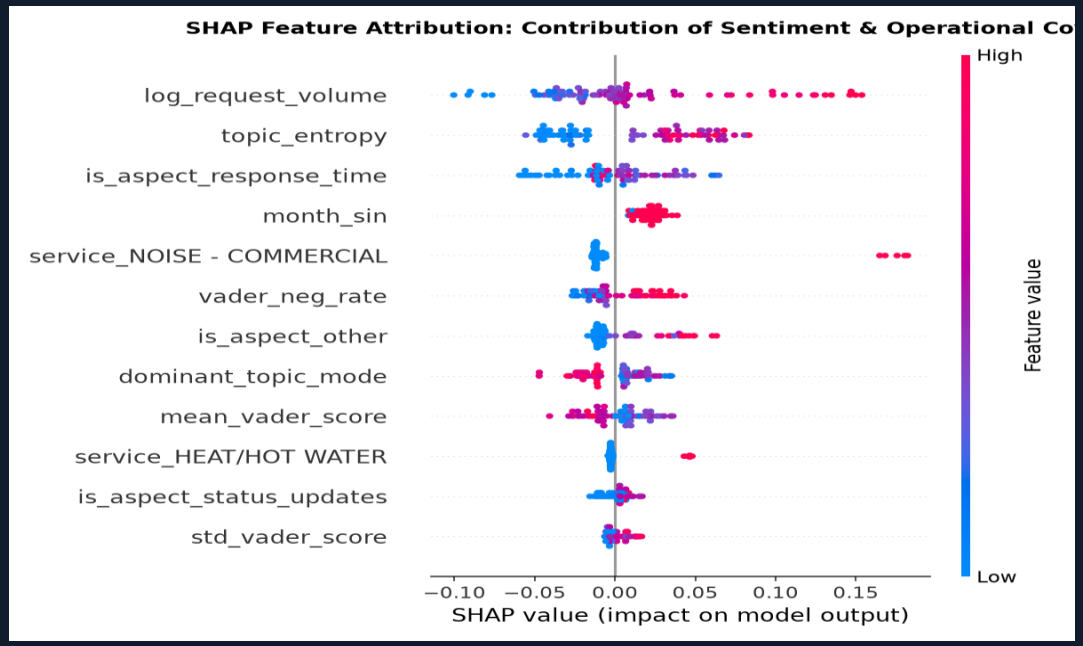
Mean ΔR^2 +0.1287 ± 0.0097
95% bootstrap CI [+0.0499, +0.2259]

Repeat rate

Mean ΔR^2 +0.0403 ± 0.0072
95% bootstrap CI [-0.0072, +0.0794]

The interval crossing zero is the key qualification.

SHAP links predictions to operational and language signals



Global SHAP feature ranking

Top contributors in the augmented delay model

- log_request_volume
- topic_entropy
- response-time aspect
- month seasonality
- VADER negative rate

Meaning

The model uses both operational pressure and civic-language signals; the dashboard exposes these signals for review rather than hiding them behind a single score.

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INTERACTIVE DASHBOARD

The final system turns the analysis into two decision-support views

Municipal Operations

- Filter by borough, service, and period
- Priority queue with risk score
- Trend and borough comparison charts
- Model evidence and SHAP view
- Downloadable priority queue

Resident View

- Plain-language KPI summary
- Historical delay and repeat trends
- Citywide service comparison
- Privacy-safe aggregate values
- No names, addresses, or ticket text

Live local link **http://localhost:8501**

Data grain: service × borough × month

IMPLEMENTATION STATUS

All major pipeline phases are implemented with inspectable artifacts

Completed pipeline

- 01 Audit
- 02 Environment
- 03 Ingestion
- 04 EDA
- 05 NLP
- 06 Outcomes
- 07 Integration
- 08 Modeling
- 09 Transformer
- 10 Dashboard
- 11 Tests
- 12 Report

Current verification

16 / 16
pytest checks passing

108
environment dependencies verified

2024
operational out-of-time horizon

The project is submission-ready when its boundaries are stated clearly

What the evidence does not claim

- No causal effect is established.
- RSAT free-text comments are PII-suppressed.
- ABSA is a domain-transfer evaluation.
- Repeat-rate gain is not significant.
- 2024 is the local operational test horizon.

Highest-value next steps

- Complete two-annotator 311 ABSA sample.
- Report Cohen’s kappa and human macro-F1.
- Collect a longer out-of-time horizon.
- Validate transfer on another 311 city.
- Monitor drift before deployment.

Academic position Strong end-to-end engineering and reproducibility; conclusions are calibrated to the available evidence.

FINAL TAKEAWAY

Municipal service quality is more than closure speed.

This project demonstrates a reproducible way to connect citizen dissatisfaction, civic text signals, and operational outcomes — with transparent limits and an actionable dashboard.

Key result	**Significant incremental evidence for resolution-delay prediction; repeat-rate evidence remains a research opportunity.**

Thank you • Questions and discussion

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